

Q26 — Detailed Solution

Q26

Question: If α and β are zeroes of the polynomial $p(x) = 2x^2 - 5x + 7$, find a quadratic polynomial whose zeroes are $2\alpha + 3\beta$ and $3\alpha + 2\beta$.

Step 1: Find $\alpha + \beta$ and $\alpha\beta$

For a quadratic polynomial $ax^2 + bx + c$, if its zeroes are α and β :

$$\alpha + \beta = -b/a$$

$$\alpha\beta = c/a$$

Here, $p(x) = 2x^2 - 5x + 7$, so $a = 2$, $b = -5$ and $c = 7$.

Therefore,

$$\alpha + \beta = -(-5)/2 = 5/2$$

$$\alpha\beta = 7/2 = 7/2$$

Step 2: Name the new zeroes

Let the new zeroes be r and s .

$$r = 2\alpha + 3\beta$$

$$s = 3\alpha + 2\beta$$

For the required quadratic polynomial, we need $r + s$ and rs .

Step 3: Find the sum of the new zeroes

$$r + s = (2\alpha + 3\beta) + (3\alpha + 2\beta)$$

$$= 5\alpha + 5\beta$$

$$= 5(\alpha + \beta)$$

$$= 5(5/2)$$

$$r + s = 25/2$$

Step 4: Find the product of the new zeroes

$$rs = (2\alpha + 3\beta)(3\alpha + 2\beta)$$

$$= 6\alpha^2 + 4\alpha\beta + 9\alpha\beta + 6\beta^2$$

$$= 6\alpha^2 + 13\alpha\beta + 6\beta^2$$

$$= 6(\alpha^2 + \beta^2) + 13\alpha\beta$$

Step 5: Find $\alpha^2 + \beta^2$

Use the identity:

$$(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$$

Therefore,

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (5/2)^2 - 2(7/2)$$

$$= 25/4 - 7$$

$$= 25/4 - 28/4$$

$$\alpha^2 + \beta^2 = -3/4$$

Step 6: Substitute into rs

$$\begin{aligned}
 rs &= 6(\alpha^2 + \beta^2) + 13\alpha\beta \\
 &= 6(-3/4) + 13(7/2) \\
 &= -9/2 + 91/2 \\
 &= 82/2 \\
 \mathbf{rs} &= \mathbf{41}
 \end{aligned}$$

Step 7: Form the new quadratic polynomial

For roots r and s, the quadratic polynomial is:

$$\begin{aligned}
 x^2 - (r + s)x + rs \\
 = x^2 - (25/2)x + 41
 \end{aligned}$$

Multiplying throughout by 2 to remove the fraction:

$$\mathbf{2x^2 - 25x + 82}$$

■ **Final Answer:**

$$\mathbf{2x^2 - 25x + 82}$$

Quick exam method

Given $2x^2 - 5x + 7$:

$$\alpha + \beta = 5/2, \quad \alpha\beta = 7/2$$

Let $r = 2\alpha + 3\beta$ and $s = 3\alpha + 2\beta$.

$$r + s = 5(\alpha + \beta) = 25/2$$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25/4 - 7 = -3/4$$

$$rs = 6(\alpha^2 + \beta^2) + 13\alpha\beta = 6(-3/4) + 13(7/2) = 41$$

$$\text{Required polynomial} = x^2 - (25/2)x + 41$$

Multiplying by 2:

$$\mathbf{2x^2 - 25x + 82}$$